

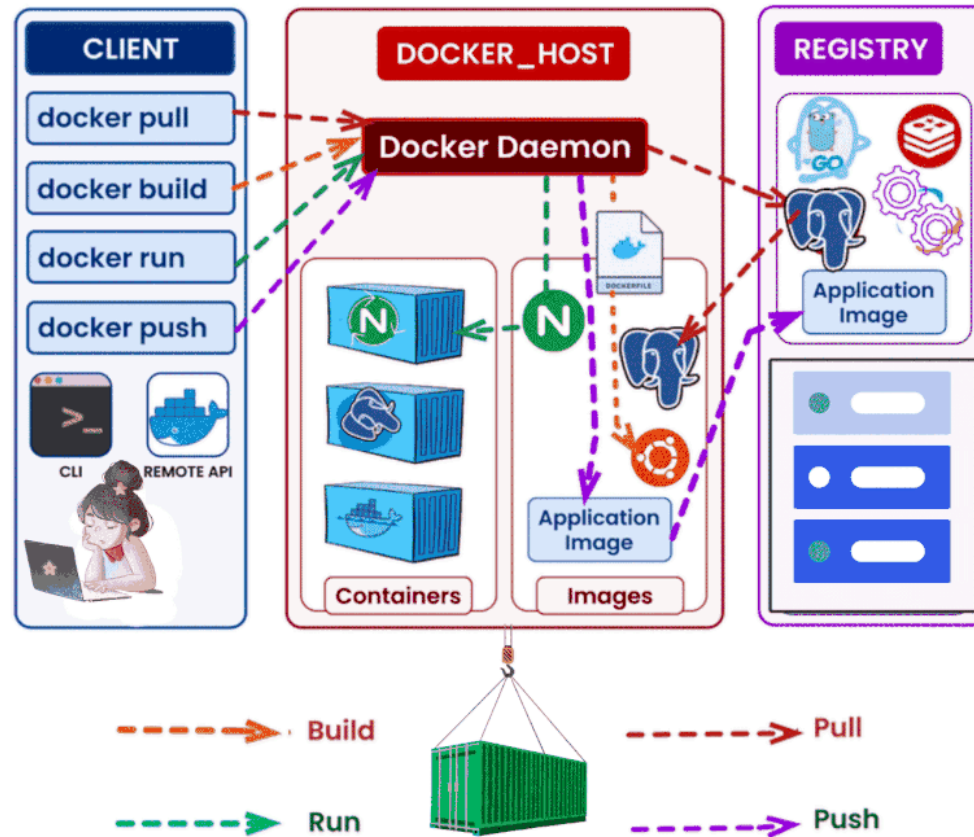
Fullstack Development

Docker

Content

- Docker recap
- Docker build
- Docker vulnerabilitiy
- Docker Swarm

DOCKER ARCHITECTURE



Docker Tutorial

Another Todo App

Awesome Compose

Docker vulnerability

Generally fall into 3 categories:

- Outdated container runtime or engine software (breakouts)
- Vulnerable 3rd-party application dependencies (image & libs)
- System misconfigurations (`:latest` tag, mount `/var/run/docker.sock`)

Remediation & Practice

- Integrate (automated) docker image security scanners
 - `trivy`, `Docker Scout`
 - [Visscan Project](#)
 - `Semgrep` : scans source code from git repository
 - `trivy` : if `Dockerfile` is available, build and scans docker image

Docker compose scaling

- The process of horizontally expanding/shrinking your app by running **multiple instances** of the same containerized service
- By default, `docker compose` spins up exactly one container per service

Docker compose scaling - How to

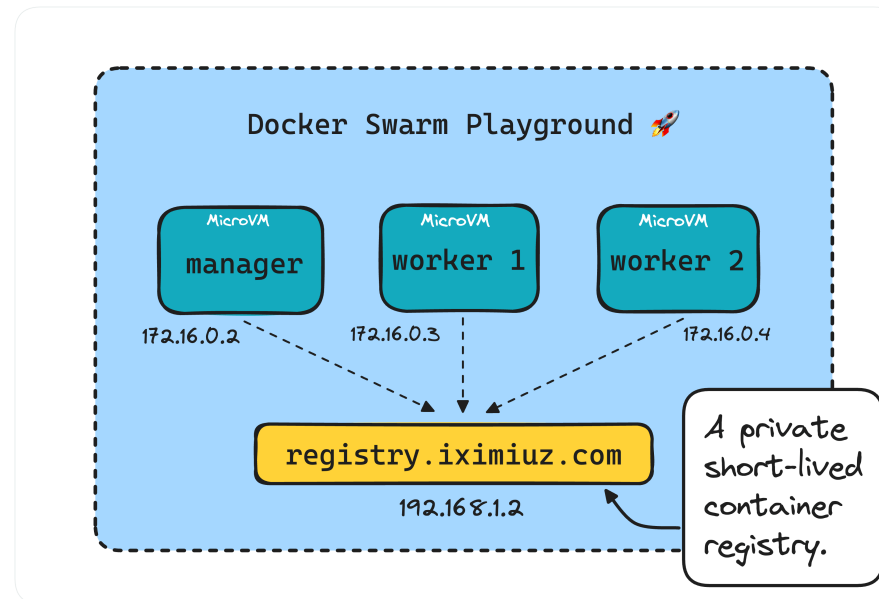
- No **static host ports** (like `"8080:80"`)
 - Instead expose only container port (`"80"`)
 - Docker assigns random host ports automatically
- No **custom container names** (like `container_name: my_app`)
 - Docker manages names by adding suffixes (`web-1` , `web-2`) when scaling
- Requires a **load balancer** (`nginx` , `caddy` , `traefik`)

Docker compose scaling - example

Demo

Docker Swarm

- Docker's native **container orchestration tool**
- Manages a cluster of multiple Docker engines/hosts as a single virtual system
 - Control deployment, scaling, networking, HA



Docker Swarm Tutorial

- [Swarm Tutorial](#)
- [Docker Swarm Playground](#)

